



VCS MONITORING REPORT

ZANO SOLAR POWER

Project title	<i>Zano Solar Power</i>
Project ID	4389
Monitoring period	<i>24-October-2023 – 31-December-2024</i>
Original date of issue	<i>06-February-2025</i>
Most recent date of issue	<i>22-May--2025</i>
Version	04
VCS Standard Version	4.7
Prepared by	<i>AERA Group</i>

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PROJECT DETAILS

1.1 Summary Description of the Implementation Status of the Project

The project consists of the construction and operation of a greenfield solar power plant, Zano for 24MWc. Zano is located near the town of Tenkodogo, in Bougou region, Center- Eastern Burkina Faso and is operated by Quadran Burkina Faso. It involves the setting up of photovoltaic (PV) panels for a total installed capacity of 24MWc which captures solar energy and produce electricity exported to the West African Power Pool through the national grid.

Access to clean and inexpensive energy is a priority of Burkina Faso's Ministry of Energy, which aims to increase the national and rural electrification rates by 45% and 19% respectively by 2025¹. To this end, the Government of Burkina Faso has signed a concession agreement with Quadran International (now rebranded Qair), Syscom Network (the sponsors) on one hand, and Quadran Burkina Faso on the other hand (the project SPV) to build and operate Zano solar power plant during 25 years.

The project is a type I project activity under sectoral scope: 1, Energy Industries (renewable and non-renewable sources). The methodology ACM0002 (Version 21.0) is applied since its purpose is the installation of a new grid-connected renewable power plant (Greenfield) as the “electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants.” Thus the project will generate emissions reductions by producing clean energy, that will save GHG emissions, by replacing the fossil fuel energy which is the baseline scenario in Burkina Faso.

The average power generation of the project for the next 7 years is estimated at 44 GWh per year, resulting in emissions reductions of 25,589 tCO₂e emission reductions per year and 179,124 tCO₂e of emission reductions over the 7 years crediting period. For this first monitoring period, the total emission reduction is 27,097 tCO₂e.

The main technical elements of the plant are the following:

- 43 960 Modules: LONGI LR5-72HPH- 545M/550M ;
- 396 Trackers: PVH Axone-DUO
- 88 Inverters : SUNGROW250HX
- 3 Transformers station: SUNGROW MVS6750 – 33KV/0.8KV

Through the National Economic and Social Development Plan (PNDES 2016-2020), the government of Burkina Faso plans to fill the energy gap. Indeed, the latter underlines its ambition to increase the share of renewable energies in the energy mix to promote the country's self-sufficiency.

Indeed, the solar potential of the Burkinabe territory allows SONABEL to consider a unique solution to the energy deficit. In fact, the country has a solar irradiation of about 5.5%. kilowatt hours per square

¹ <https://faolex.fao.org/docs/pdf/bkf219602.pdf> p.11

meter per day. This results in a better match between the production and the maximum demand on the grid, the hybrid solution will bring also additional support to the network operator.

Table 1: Implementation timeline

PPA signature	20-November-2019
Producing License	06-April-2020
Environmental Assessment Study and stakeholder meetings	16-June-2020
Environmental clearance certificate	07-August-2020
Facility agreement	23-August-2021
Start date of project, day when electricity is fed into the grid and invoicing starts	24-October-2023 (continuous operation since then)
VCS validation	10-July-2024
First VCS monitoring period	24-October-2023 – 31-December-2024

Prior to the implementation of the project, the existing scenario consisted in power generation and imports from SONABEL existing grid mix, mainly composed of fossil-fuel (87%) and hydropower sources (13%).

FAR' from validation:

FAR01: Verifying VVB to check the commissioning certificate to verify the commissioning date of the project activity at the time of verification:

- please see the “30-days period end notification letter”. The first invoices are also available for cross checking of the date. COD certificate from SONABEL not available yet.

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Validation	24-March-2023 – 10-July-2024	VCS	Applus	15 months
Verification	24-October-2023 – 31-December-2024	VCS	RE Carbon	14 months and 6 days

1.3 Sectoral Scope and Project Type

Sectoral scope ²	1 – Energy industries (renewable / non renewable sources)
Project activity type	Renewable energy (solar power)

² Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: <https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes>

1.4 Project Proponent

Organization name	Quadran Burkina Faso
Contact person	Bruno Baritello
Title	Managing director / Country Director
Address	Projet ZACA Ouagadougou, BURKINA FASO
Telephone	+226 65 89 92 75 / 25 33 33 77
Email	b.baritello@qair.energy

1.5 Other Entities Involved in the Project

Organization name	AERA Group
Role in the project	Carbon consultant
Contact person	Aurélie Lepage
Title	CEO
Address	28 cours Albert 1er 75008 Paris
Telephone	+33 1 42 18 02 02
Email	a.lepage@aera-group.fr

1.6 Project Start Date

Project start date	24-October-2023
Justification	Commissioning of the plant, thus the date when the project started reducing emissions.

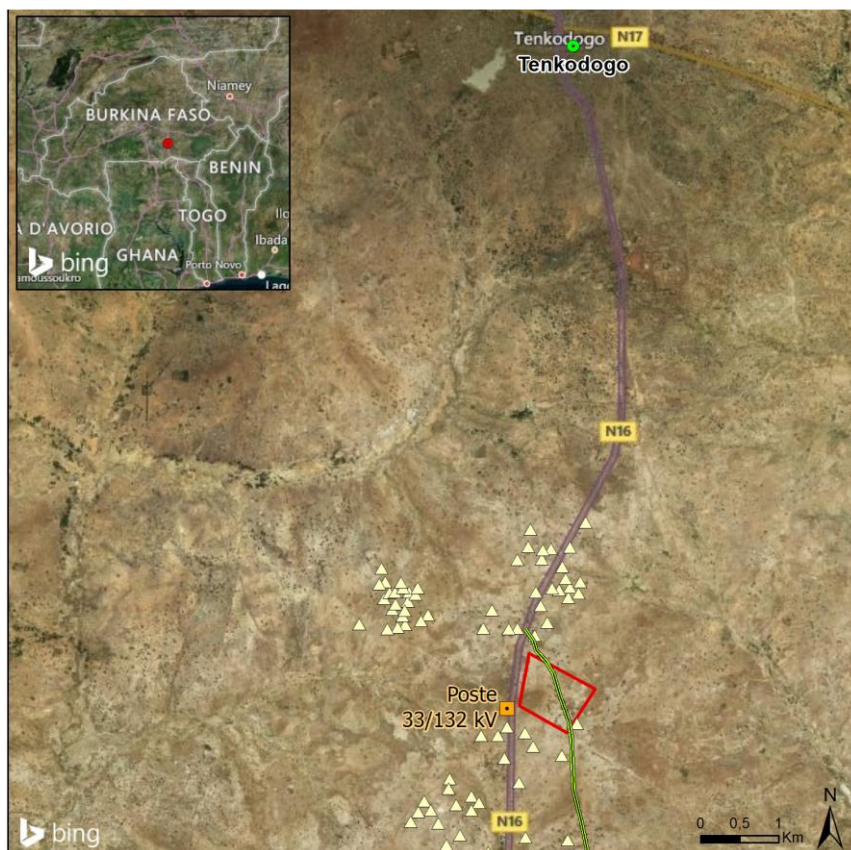
1.7 Project Crediting Period

Crediting period	☒ Seven years, twice renewable ☐ Ten years, fixed ☐ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)
Start and end date of first or fixed crediting period	24-October -2023 – 23-October-2030

1.8 Project Location

The Zano site is located about 140 km southeast of Ouagadougou in the locality of Zano, located in the department of Tenkodogo, Boulgou province, Centre-East region, Burkina Faso. The site covers an area of 63.56 hectares in flat surroundings. The direct right-of-way of the Project is

shown on the following map. It should be noted that no buildings are on the Project site, but a sand track crosses it.



GPS coordinates: 11°42'13248"N ; - 0°07'26.4W



Source: Google maps

1.9 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	ACM0002	Grid-connected electricity generation from renewable sources"	21.0
Standardized baseline	ASB0034-2021	Grid emission factor for West African Power Pool	1.0
Methodological tool	Tool 01	Tool for the demonstration and assessment of additionality	7.0.0
Methodological tool	Tool 05	Baseline, project and / or leakage emissions from electricity consumption and monitoring of electricity generation	3.0
Methodological tool	Tool 07	Tool to calculate the emission factor for an electricity system.	7.0
Methodological tool	Tool 24	Common practice.	0.3.1

1.10 Double Counting and Participation under Other GHG Programs

1.10.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

If yes, provide required evidence of no double issuance as outlined by the VCS Standard.

1.10.2 Registration in Other GHG Programs

Was the project registered or seeking registration under any other GHG programs?

☐ Yes ☒ No

If yes, provide the registration number and all relevant details including the date of project inactivity in the other GHG program.³

1.11 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.11.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

If yes, provide all required evidence of no double claiming as outlined by the VCS Standard.

1.11.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

If yes, provide all required evidence of no double claiming as outlined by the VCS Standard.

1.11.3 Supply Chain (Scope 3) Emissions

Do the project activities affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

³ The requirement to provide the date of project inactivity only applies to projects which request registration or crediting period renewal under the VCS Program on or after 1 January 2025.

☐ Yes ☒ No

If yes:

Is the project proponent(s) or authorized representative a buyer or seller of the product(s) (goods or services) that are part of a supply chain?

☐ Yes ☐ No

If yes:

Has the project proponent(s) or authorized representative posted a public statement on their website saying, "Carbon credits may be issued through the Verified Carbon Standard project [project ID] for the greenhouse gas emission reductions or removals associated with [project proponent or authorized representative organization name(s)] [name of product(s) whose emissions footprint is changed by the project activities]."

☐ Yes ☐ No

If yes to all:

Provide evidence of the public statement. Evidence must be provided in this section or in an appendix.

1.12 Sustainable Development Contributions

The project produced renewable energy, feeding the grid with energy that would otherwise have been produced by thermal plants (SDG 7). To do so, an average of 57 person were employed by the project every month, out of which an average of 55 are local people, and full time employed (only an average of 33 are permanent workers). (SDG 8)

By producing renewable energy from solar panels, the project reduced CO₂ emissions. (SDG 13)

Table 2: Sustainable Development Contributions

Row number	SDG target	SDG indicator	Net impact on SDG indicator	Current project contributions	Contributions over project lifetime
1)	7.2	7.2.1 Renewable energy share in the total final energy consumption	Implemented activities to increase	The project produced 47,291 MWh of renewable energy during this period.	The project produced 47,291 MWh of renewable energy over its lifetime.
2)	8.5	Total number of jobs created	Implemented activities to increase	The project hired an average of 57 persons full time per month. Average over monitoring period: 55 local people – 24 temporary workers – 33 full time permanent workers	The project hired an average of 57 persons full time per month
3)	13.0	Tonnes of greenhouse gas emissions avoided or removed	Implemented activities to increase	By producing energy through solar power, the project has reduced 27,097 tCO _{2e}	Prevented the release of 27,097 tonnes of carbon into the atmosphere over its lifetime

1.13 Commercially Sensitive Information

Not applicable as there is no commercially sensitive information.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification	Not applicable, the stakeholders were identified during ESIA, and the list of project affected person (PAP) is available. The project continues working with the same group of PAP since the beginning. (107 person).
Legal or customary tenure/access rights	SONABEL, the national electricity company bought the land back in 2016. Previous owners were allowed to cultivate the land until the start of construction. They are the 107 identified PAP for whom a resettlement plan was made.
Stakeholder diversity and changes over time	The PAP's are either the owners of the land either the farmers. They won't change over time, and other stakeholders are the inhabitants of the surroundings, mainly farmers.
Expected changes in well-being	The main and only change is that the farmers won't be able to cultivate the land bought for the project. Though they have known this for years, and most of them have other land to cultivate also. The project will bring new jobs for local people, and financial help for schools and other projects.
Location of stakeholders	The stakeholders live in the surrounding of the project, in the village of Zano, which is dispersed. No changes to be expected.
Location of resources	The land of the project was cultivated, but is not anymore. All 107 farmers have been compensated and are getting support from the project.

2.1.2 Stakeholder Consultation and Ongoing Communication

Ongoing consultation	The project held 8 meetings with local stakeholders, and 43 individual consultation in 2024.
Date(s) of stakeholder consultation	October – November - December 2023 January 24 individual consultation + meeting with local authority April 24 September 24 – individual + collective consultation. – October 24 – November 24
Communication of monitored results	The meetings are regular and the results can easily be communicated from one meeting to another.
Consultation records	annual reports
Stakeholder input	No feedback was gathered in the reports.

2.1.3 Free, Prior, and Informed Consent

Consent	As per EIA report, none of the different stakeholders met rejected the project. However, each category of actors raised concerns related to their position in relation to the project. This is why the main concerns raised by local authorities and technical services revolve around compliance with the regulations and procedures in force in Burkina Faso. As for local elected officials and the populations they represent, they not only expressed the wish to be able to fully benefit from the benefits of the project, but also took charge of this expectation by negotiating conditions of compensation and support from the promoter of the project.
Outcome of FPIC	The project received environmental Clearance Certificate on 31-August-2020. The project did not encroach on land, relocated people without consent, and did not force physical or economic displacement

2.1.4 Grievance Redress Procedure

Grievances received	Resolution and outcome
Mr YAMNEDA Lassane CVD of ZANO requested verification of the household status of PAPs	We hereby inform you that PAPs are eligible to constitute households in their own right. Therefore, in addition to the

N° 44 Kéré Lydie, N° 76 - SORGHO Assiratu and N° 81 - SORGHO Sahibata, explaining that they do not benefit from the projects carried out per household even though they represent households in their own right.	cereals, they will also benefit from an electricity connection as part of the energy access project. Closing date: 20-October-2023
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2.1.5 Public Comments

Summary of comments received	Actions taken
<i>No comments received during this monitoring period</i>	<i>N/A</i>
...	

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

Qair international developed several renewable energy projects world wide and is implemented in multiple countries. Their ethical charter is very clear on different subject such as management, sexual and any other type of harassment, gender equality, environment, etc.

2.2.2 Risk assessment

	Risk identified	Mitigation or preventative measure(s) taken
Natural and human-induced risks to stakeholders' wellbeing	No risk identified	As per ESIA report there are no risk to stakeholders' wellbeing
Risks to stakeholder participation	No risk identified	The stakeholders are regularly invited to group or individual consultation
Working conditions	Health and safety of workers	Implement the Health, Safety and Security plan for workers and train all workers : ■ Ensure the wearing of appropriate PPE, particularly when handling hazardous materials or waste and machinery

		■ Clearly identify the areas of storage of hazardous materials or waste ■ Allow only workers trained and certified to work on electrical equipment ■ To have the systems of stop automatic switches placed in the stations allowing automatic shutdown and warning of the control center in case of unusual overheating (fire) ■ The emergency vehicles will use the existing access roads allowing them to access the closest points of the site ■ To provide means of fire-fighting equipment (appropriate extinguishers) placed at the transformers
Safety of women and girls	No risk identified	The company has an equal chance policy and will hire equally a man or a woman for the same task
Safety of minority and marginalized groups, including children	No risk identified	The safety of minority / marginalized / children was not identified as a risk
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)	Management of waste domestic and industrial	Implement the waste management plan that will identify waste disposal channels for each type of waste that will comply with Burkinabe regulations and international best practices: ■ In particular, the issue of recycling photovoltaic panels will be evaluated. ■ The control measures will be recorded in a Waste Management Plan ■ Waste transport vehicles will be in good condition, maintained, and operators trained ■ Subcontractors used will be approved by the operator ■ One or more waste disposal facilities will be selected

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

	Risks identified ⁴	Mitigation or preventative measure(s) taken
Discrimination	No risk identified	As per grievance mechanism available as well as HSE reports, no discrimination nor sexual harassment occurred during this monitoring period
Sexual harassment	No risk identified	As per grievance mechanism available as well as HSE reports, no discrimination nor sexual harassment occurred during this monitoring period
Gender equity in labor and work	No risk identified	Qair has a very clear ethical charter on the subject
Forced labor	No risk identified	The project aligns with the labour convention and labour rights of Burkina Faso, which is a member of the International Labour Organisation (ILO) since 04.11.1960. ⁵
Child labor	No risk identified	The project aligns with the labour convention and labour rights of Burkina Faso, which is a member of the International Labour Organisation (ILO) since 04.11.1960. ⁶
Human trafficking	No risk identified	The project aligns with the labour convention and labour rights of Burkina Faso, which is a member of the International Labour Organisation (ILO) since 04.11.1960. ⁷

2.3.2 Human Rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	Quadrant Burkina Faso employs local population for the plant management and the project aligns with the labour convention and labour rights of Burkina Faso, which is a member of the International Labour Organisation (ILO) since 04.11.1960 and signed the following fundamental conventions :

⁴ The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

⁵ https://www.ilo.org/dyn/normlex/en/f?p=1000:11110:0::NO:11110:P11110_COUNTRY_ID:103013

⁶ https://www.ilo.org/dyn/normlex/en/f?p=1000:11110:0::NO:11110:P11110_COUNTRY_ID:103013

⁷ https://www.ilo.org/dyn/normlex/en/f?p=1000:11110:0::NO:11110:P11110_COUNTRY_ID:103013

	1. Freedom of Association and Protection of the Right to Organise Convention, 1948 (No. 87); 2. Right to Organise and Collective Bargaining Convention, 1949 (No. 98); 3. Forced Labour Convention, 1930 (No. 29); 4. Abolition of Forced Labour Convention, 1957 (No. 105); 5. Minimum Age Convention, 1973 (No. 138); 6. Worst Forms of Child Labour Convention, 1999 (No. 182); 7. Equal Remuneration Convention, 1951 (No. 100); and, 8. Discrimination (Employment and Occupation) Convention, 1958 (No. 111);
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2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Mitigation(s) or preventative measure taken
No risk identified	Not applicable as there was no tangible or intangible cultural heritage in the area.

2.3.4 Property Rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	No change in land acquired since agreement between stakeholders and PP during project's implementation

2.3.5 Benefit Sharing

Summary of the benefit sharing plan	<i>No benefit sharing plan. The communities benefit from the social responsibility plan and PRME of the project</i>
Benefit sharing during the monitoring period	<i>No benefit sharing plan. The communities benefit from the social responsibility plan and PRME of the project</i>

2.4 Ecosystem Health

	Risk identified	Mitigation or preventative measure(s) taken during the monitoring period
Impacts on biodiversity and ecosystems	No risk identified	No risk identified as per EIA report.

Soil degradation and soil erosion	No risk identified	No soil degradation during this monitoring period as per latest HSE report.
Water consumption and stress	No risk identified	No risk related to water consumption was identified for operation phase as per EIA report

2.4.1 Rare, Threatened, and Endangered species

Species or habitat	Not identified by ESIA as a risk, as there are no endangered species on the project site.	
Areas needed for habitat connectivity	N/A	
...	...	

	Risks identified	Mitigation or preventative measure(s) taken
Habitats for rare, threatened, and endangered species	No risk identified	As per ESIA report, no impact on habitats for threatened and endangered species.
Areas for habitat connectivity	No risk identified	N/A

2.4.2 Introduction of species

N/A

2.4.3 Ecosystem conversion

N/A

3 IMPLEMENTATION STATUS

3.1 Implementation Status of the Project Activity

After obtaining the environmental clearance, the producing license and the power purchase agreement late in 2019 and 2020, the project could finally achieve construction during late 2023. The validation of the project occurred at the same time than construction.

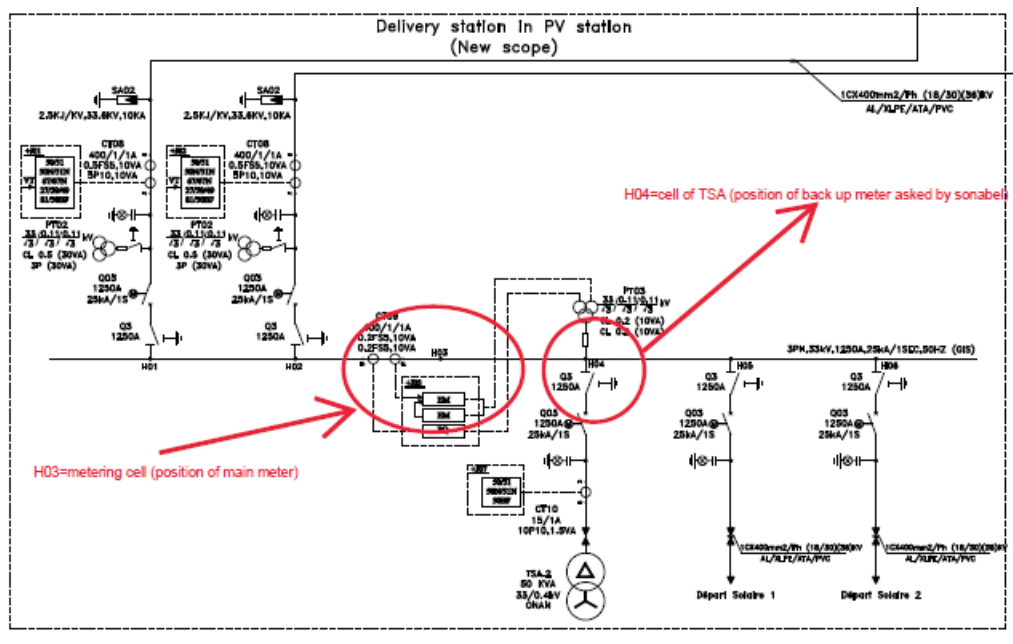
After a month of testing, the project started selling electricity to the grid starting 24-October-2023. Though selling the electricity half-price to SONABEL should have happened for only one month after 24-October-2023 this period lasted longer, and official commercial operation day was thus delayed by several month. From an emission reductions' point of view though, the project has been feeding and invoicing renewable electricity since 24-October-2023 for a total of 47,291 MWh and 27,097 tCO₂e reduced over this first monitoring period.

The project didn't encounter any major breakdowns and the regular maintenance occurred without shutting down the entire plant.

The main technical elements of the plant are the following:

- 43 960 Modules: LONGI LR5-72HPH- 545M/550M ;
- 396 Trackers: PVH Axone-DUO
- 88 Inverters : SUNGROW250HX
- 3 Transformers station: SUNGROW MVS6750 - 33KV/0.8KV

The meters installed are the following:

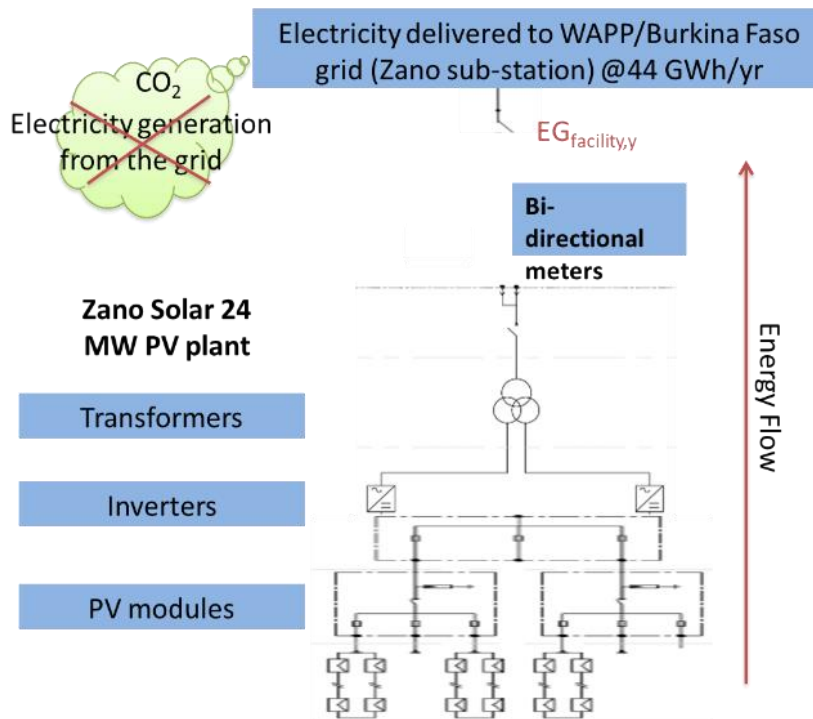


Main meter : Prometer100 SN: WP050493

Backup meter : Landis+Gyr E650 SN: 59383221

Accuracy class: 0.2S ($\pm 0.2\%$)

The backup meter was changed on 12-March-2024 (as per report provided), thus the new backup meter is now Prometer100 SN: WP060048 (accuracy class 0.2S)



3.2 Deviations

3.2.1 Methodology Deviations

N/A as no methodology deviation applied during this monitoring report.

3.2.2 Project Description Deviations

N/A as no project description deviation occurred during this monitoring report.

3.3 Grouped Projects

N/A as not a grouped project.

3.4 Baseline Reassessment

Did the project undergo baseline reassessment during the monitoring period?

☐ Yes

☒ No

4 DATA AND PARAMETERS

4.1 Data and Parameters Available at Validation

Data / Parameter	$EF_{grid,CM,y}$
Data unit	tCO ₂ /MWh
Description	Combined margin CO ₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system”
Source of data	As per ASB0034-2021 https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20210325112041972/ASB0034-2021.pdf
Value applied	0.573
Justification of choice of data or description of measurement methods and procedures applied	As per the Tool 07 “Tool to calculate the emission factor for an electricity system” version 07.0
Purpose of data	Calculation of baseline emissions
Comments	-

4.2 Data and Parameters Monitored

Data / Parameter	$EG_{facility,y}$				
Data unit	MWh/monitoring period				
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid during first monitoring period				
Source of data	Measured directly with electricity meter(s) at project site. Readings done monthly both by representative of Quaran Burkina and Sonabel.				
Description of measurement methods and procedures to be applied	Electricity outputs will be electronically stored and invoiced monthly to Sonabel by Quadran Burkina Faso, after reconciliation with internal meter which readings are recorded on a record sheet by the Technical/Engineering/ Maintenance Department under the Plant Manager’s authority				
Frequency of monitoring/recording	Continuous measurement and at least monthly recording				
Value monitored	<table border="1"> <tr> <td>24-October-2023 – 31-December-2023</td><td>7,832</td></tr> <tr> <td>01-January-2024 – 31-December 2024</td><td>39,459</td></tr> </table>	24-October-2023 – 31-December-2023	7,832	01-January-2024 – 31-December 2024	39,459
24-October-2023 – 31-December-2023	7,832				
01-January-2024 – 31-December 2024	39,459				

Monitoring equipment	Bi-directional electronic electricity meters measuring the net electrical energy delivered to Sonabel with the following specifications: Main invoicing meter: Prometer100 SN : WP050493 Accuracy class: 0.2S ($\pm 0.2\%$) Backup meter: Prometer100 SN: WP060048 Accuracy class: 0.2S
QA/QC procedures to be applied	Cross check of measurement results with invoiced electricity. Calibration of the meters will be carried out after each deviation of more than $\pm 2\%$ between the 2 meters as per PPA specifications. A verification of the meters is done by a certified third party every three years, as per PPA requirements. The installation and verification of WP050493 (main meter) was done from 21 to 24th September 2023, next due verification before 24-September-2026. The installation and verification of WP060048 (back up meter) was done on 12-March-2024, next due verification before 12-March-2027. There are also two Sonabel meters installed which serve for cross checking, but not under PP's responsibility.
Purpose of the data	Calculation of baseline emissions
Calculation method	Electronic recording
Comments	The back up meter was changed on 14-March-2024. Documentation available.

4.3 Monitoring Plan

The proposed project activity monitoring plan complies with the methodology ACM0002 - Consolidated baseline methodology for grid-connected electricity generation from renewable sources version 21.0, whereby it is stated that:

“All data collected as part of monitoring should be archived electronically and be kept at least for 2 years after the end of the last crediting period”.

Therefore, the quantity of net electricity generation supplied by the project plant to the grid will be reliably monitored through calibrated electricity meters and cross-checked with sales records as per internal monitoring practices.

The quantity of emission reduced is recorded through monitoring reports, while the number of employees is recorded both by annual reports and PP's internal files.

Monitoring team and training and quality management procedures

The quantity of net electricity generation is collected by representatives of both Quadran Burkina Faso and Sonabel each month by reading all the meters (two from Quadran Burkina Faso, main and back-up and two pertaining to Sonabel, thus 3 meters for cross checking, as only Quadran's main is used for invoicing). These data can be cross checked with the SCADA and following with the invoices which is based on Quadran's main meter readings.

The implementation of a system of supervision and control (SCADA) that transfers real-time all monitoring information for off-site recording allows the operator to access and retrieve safely all Project data regardless of eventual incidents.

Meter measurements shall not exceed $\pm 2\%$ difference, beyond which dysfunctional meter will in all cases need to be identified by Sonabel and Quadran Burkina Faso, adjusted or replaced within 48 hours in accordance with manufacturer guidelines. After each deviation of more than $\pm 1\%$, a test and calibration of the meters will be carried out for each meter.. Otherwise, verification of meters is done every three years, as per PPA requirements.

Non-conformance data during uncalibrated periods will systematically be discounted by the corresponding error margin.

As emission reductions are solely based on calculation from energy produced, these calculations are made either by carbon consultant (AERA Group for this first monitoring period) within monitoring reports, or by Quadran Burkina Faso, also in monitoring reports.

The Project Director of Quadran Burkina Faso coordinates and endorses the overall responsibility for all VCS monitoring of the project, including:

- Develop, approve, execute, and improve the VCS Monitoring/Reporting Procedures;
- Organize in-house seminar to inform and train the company staff to the monitoring procedures;
- Ensure that instrumentations and devices are available and properly suited to efficiently perform the monitoring;
- Communicate and coordinate the monitoring work of all business units;
- Validate and electronically archive all monitoring data on a monthly basis throughout the crediting period (and conserve it at least for 2 further years);
- Calculate and report the emission reductions; and
- Coordinate the DOE work during the verification audit.

The Project Director might appoint a VCS coordinator, who is an internal employee of Quadran Burkina Faso, to delegate him the above specific tasks of monitoring supervision.

The Operation and Maintenance contractor, Elsewedy, consisting of plant technicians will undertake the technical actions required by the monitoring plan, under the Project Director's authority, to collect and record related data.

The meter readings are done jointly by Quadran Burkina Faso and Sonabel, every month.

The Administrative Assistant or Administrative Department will crosscheck, reconcile or consolidate data with multiple sources whenever possible. At minimum, data obtained from the electricity meters is to be crosschecked with the electricity sales invoices. The invoicing is made by the producer, i.e. Quadran Burkina Faso

Sonabel, along with SI3E is solely responsible for the selection, installation, calibration, servicing, testing and repairing of all energy meters

The GHG calculation is undertaken by the carbon consultant (Aera Group)

Trainings of employees

Although not related to monitoring, training of employees happens over the monitoring period.

During this first monitoring period the site manager and QHSE manager were successfully trained on Sungrow products. There are also some HSE trainings for workers.

5 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

Baseline emissions include only CO₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions are to be calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y} \quad (4)$$

Where:

BE_y = Baseline emissions in year y (tCO₂/yr)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EF_{grid,CM,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of “TOOL07: Tool to calculate the emission factor for an electricity system” (tCO₂/MWh)

Calculation of $EG_{PJ,y}$

Since the project activity is the installation of a new grid-connected renewable power plant at a site where no renewable power plant was operated prior to the implementation of the project activity, it verifies the case of a Greenfield renewable energy power plant of the ACM0002 methodology Version 21.0 whereby:

$$EG_{PJ,y} = EG_{facility,y} \quad (5)$$

Where:

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EG_{facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

$EG_{facility,y}$ is therefore the quantity of net electricity supplied by the project plant to SONABEL electricity grid. It is determined as a difference between (i) quantity of electricity supplied by the project plant to the grid and (ii) quantity of electricity delivered to the project plant from the grid (please refer to section B.7 for monitoring details). The methodology ACM0002 Version 21.0 assumes that all project electricity generation above baseline levels would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in $EF_{grid,CM,y}$.

$$EG_{facility,2023-2024} = 47,291 \text{ MWh}$$

Calculation of $EF_{grid,CM,y}$

Based on Standardized Baseline ASB0034-2021 Grid emission factor for West African Power Pool v1.0, the applicable grid emission factor value to calculate the emission reductions of the PV power plant project is 0.573 tCO₂/MWh).

5.2 Project Emissions

Project emissions shall be accounted by using the following equation:

$$PE_y = PE_{FF,y} + PE_{GP,y} + PE_{HP,y} \quad (2)$$

Where:

PE_y = Project emissions in year y (tCO₂e/yr)

$PE_{FF,y}$ = Project emissions from fossil fuel consumption in year y (tCO₂/yr)

$PE_{GP,y}$ = Project emissions from the operation of dry, flash steam or binary geothermal power plants in year y (t CO₂e/yr)

$PE_{HP,y}$ = Project emission from water reservoirs of hydro power plants in year y (tCO₂e/yr)

Project emissions from fossil fuel consumption ($PE_{FF,y}$)

No project emissions are expected as the project activity only involves renewable electricity generation from the solar power plant without fossil fuel consumption, and according to para 36 of ACM0002 “for all renewable energy power generation activities, emissions due to the use of fossil fuels for the backup generator can be neglected, hence $PE_{FF,y} = 0$.

Project emission from the operation of dry, flash steam or binary geothermal power plants ($PE_{GP,y}$)

Project is a solar power plant hence inapplicable and $PE_{GP,y} = 0$.

Emissions from water reservoirs of hydro power plants ($PE_{HP,y}$)

Project is a solar power plant hence inapplicable and $PE_{HP,y} = 0$

5.3 Leakage Emissions

No leakage is accounted for in the methodology.

5.4 GHG Emission Reductions and Carbon Dioxide Removals

According to the approved methodology ACM0002 ver.21.0, emission reductions are calculated as follows:

$$ER_y = BE_y - PE_y$$

Where:

ER_y = Emission reductions in year y (t CO₂e)

BE_y = Baseline emissions in year y (t CO₂)

PE_y = Project emissions in year y (t CO₂e)

Vintage period	Baseline emissions (tCO ₂ e)	Project emissions (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Reduction VCUs (tCO ₂ e)	Removal VCUs (tCO ₂ e)	Total VCUs (tCO ₂ e)
24-Oct-2023 to 31-Dec-2023	4,487	0	0	4,487	0	4,487
01-Jan-2024 to 31-Dec-2024	22,610	0	0	22,610	0	22,610

Total	27,097	0	0	27,097	0	27,097
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Vintage period	Ex-ante estimated reductions/removals	Achieved reductions/removals	Percent difference	Explanation for the difference
24-Oct-2023 to 31-Dec-2023	4,936	4,487	-10%	The difference is due to ex-ante solar irradiation estimates versus real life conditions.
01-Jan-2024 to 31-Dec-2024	25,967	22,610	-13%	The difference is due to ex-ante solar irradiation estimates versus real life conditions.
Total	30,903	27,097	-12.5%	The difference is due to ex-ante solar irradiation estimates versus real life conditions.

APPENDIX 1: COMMERCIALY SENSITIVE INFORMATION

N/A as no commercially sensitive information included